UNIVERSAL PLASTIC SHREDDER V2.0

Electrical Control & Power System

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The Problem

Ichor Systems generates large volumes of thick rigid plastic (PVC, PP, LDPE) during semiconductor equipment manufacturing. Off-the-shelf shredders are either too small for the material (0.25–0.5 in. sheets, 4–12 in. wide) or priced far above the sponsor's per-line budget. The EE team designed and built the electrical control and power system for a new industrial-grade plastic shredder. The mechanical structure and cutting blades are delivered by a separate ME team.

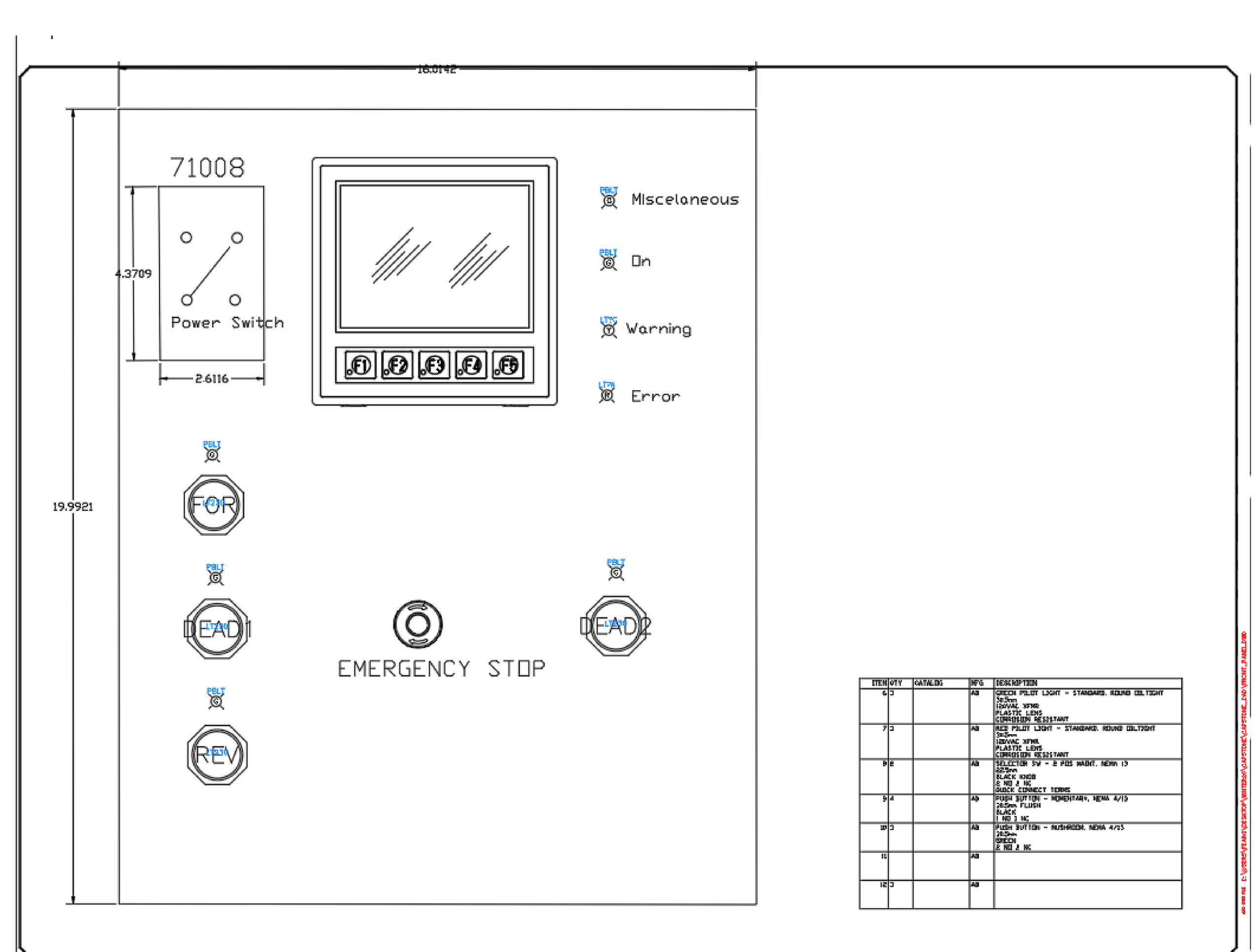
Requirements

Must:

- Shred plastic to ∼1 in. pieces.
- Drive a 3–5 HP motor at 70–90 RPM through a controllable VFD.
- Follow NFPA 70 (NEC); include E-stop, overload, and one safety interlock.
- Provide an HMI for operator control and status.
- Stay within the $1,000 EE-team budget.

Should: status / fault lights, NEMA enclosure, labeled controls.
May: read-only wireless monitoring, data logging, capacitive-touch safety.

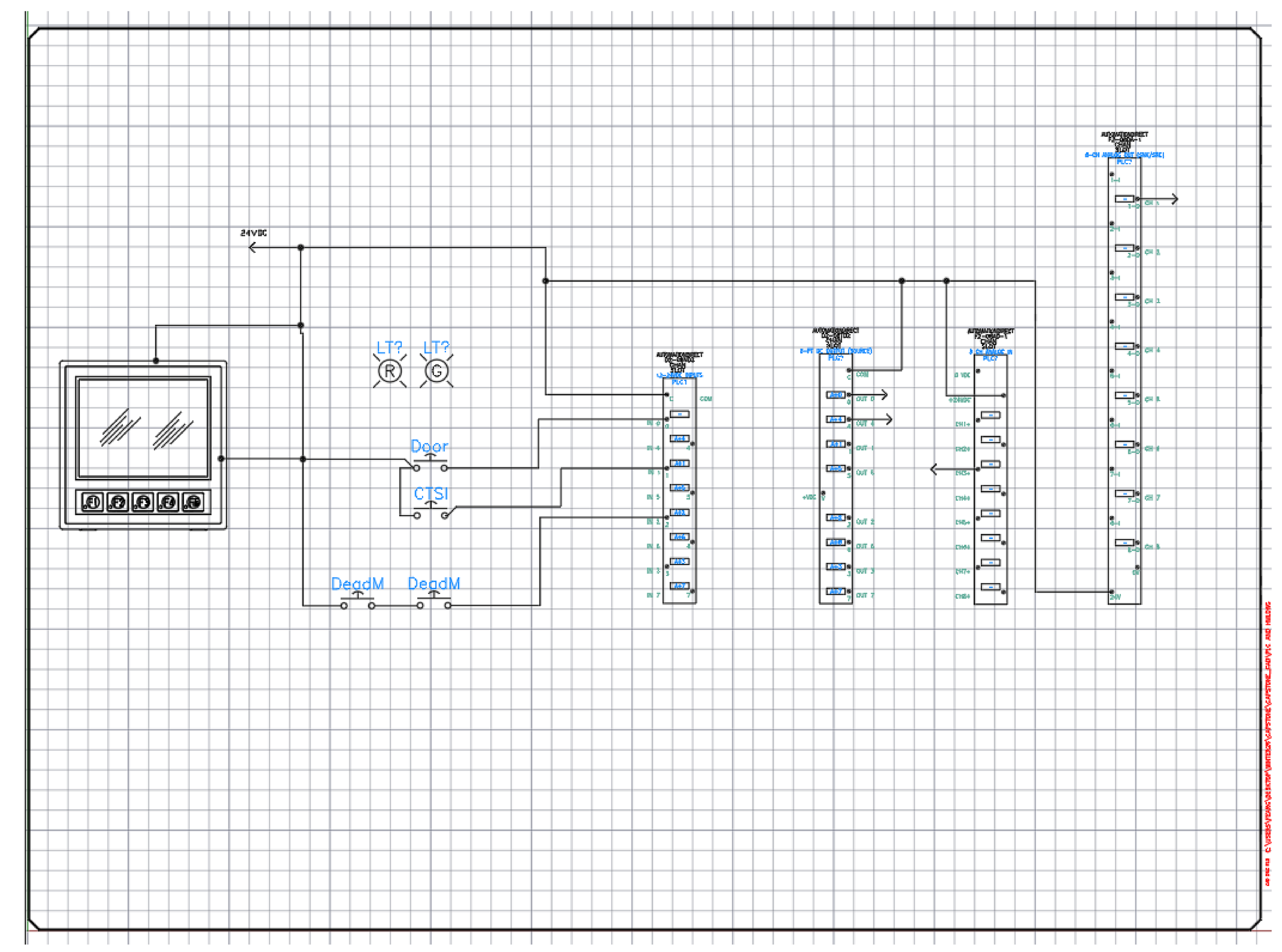
System Architecture



As-built operator control panel: HMI, power switch, forward / reverse / deadman pushbuttons, E-stop, and status indicators.

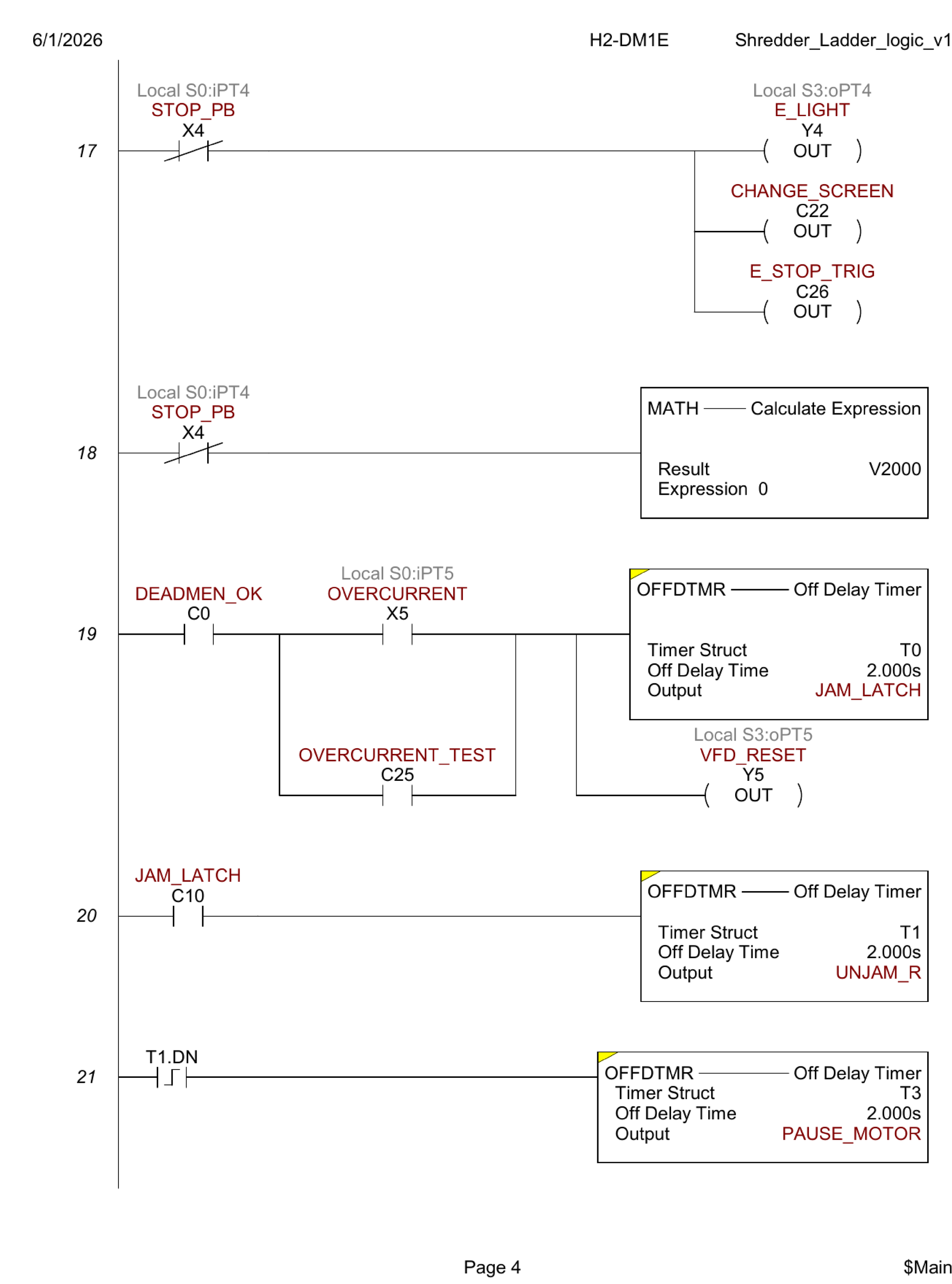
The control system is built end-to-end on AutomationDirect industrial hardware – no hobby-grade microcontrollers are in the safety or control path. A PowerTec 71008 disconnect feeds a Mean Well NDR-480-24 24 VDC supply, a Mitsubishi SD-N35 contactor (E-stop hardwired in series with its coil), and a Huanyang HY02D211B-T VFD. The VFD drives a 3 HP bench motor (GE 5KE182BC205B) over U/V/W via blue, white, and red 4 mm banana sockets.

PLC Control



DirectLOGIC 205 rack with H2-DM1E CPU and the five I/O modules.

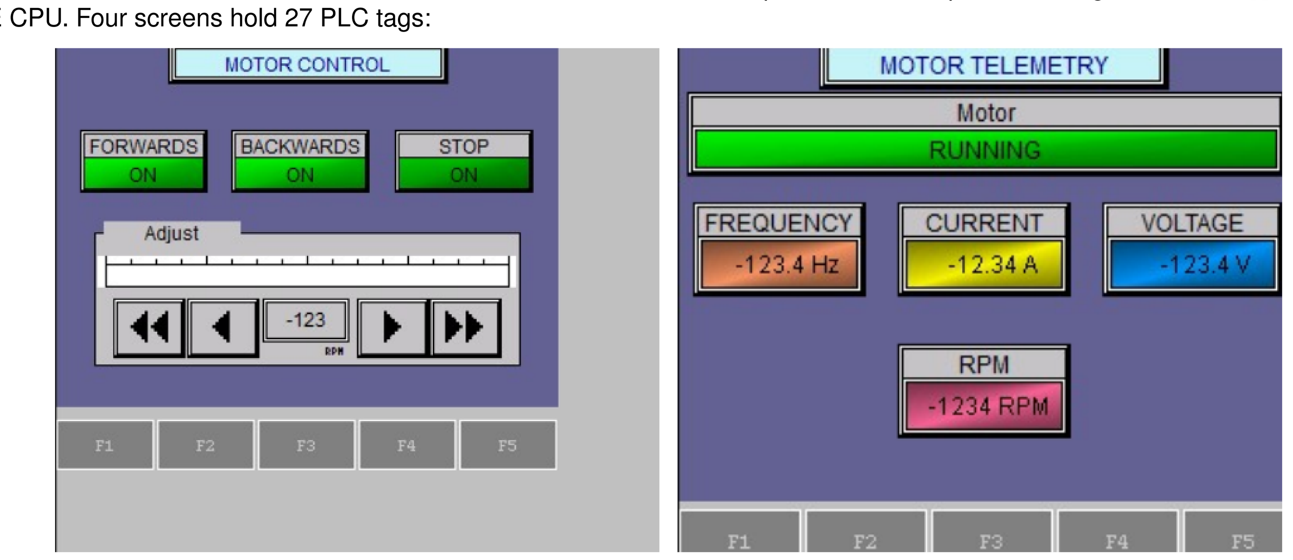
The PLC is the system controller. A DirectLOGIC 205 D2-09B-1 rack holds the H2-DM1E CPU and five I/O modules: `D2-08ND3` (8 DC inputs for pushbuttons, E-stop monitor, VFD fault, capacitive touch), `F2-08AD-1` (analog current input for VFD VO motor current), `F2-04RTD` (motor temperature, future), `D2-08TR` (8 relay outputs for VFD RUN_FWD / RUN_REV / RESET and panel LEDs), and `F2-08DA-2` (0–10 V analog output to the VFD VI speed reference). The 31-rung Do-more program implements a deterministic state machine plus a 3-strike jam recovery sequence and a hard-lockout path.



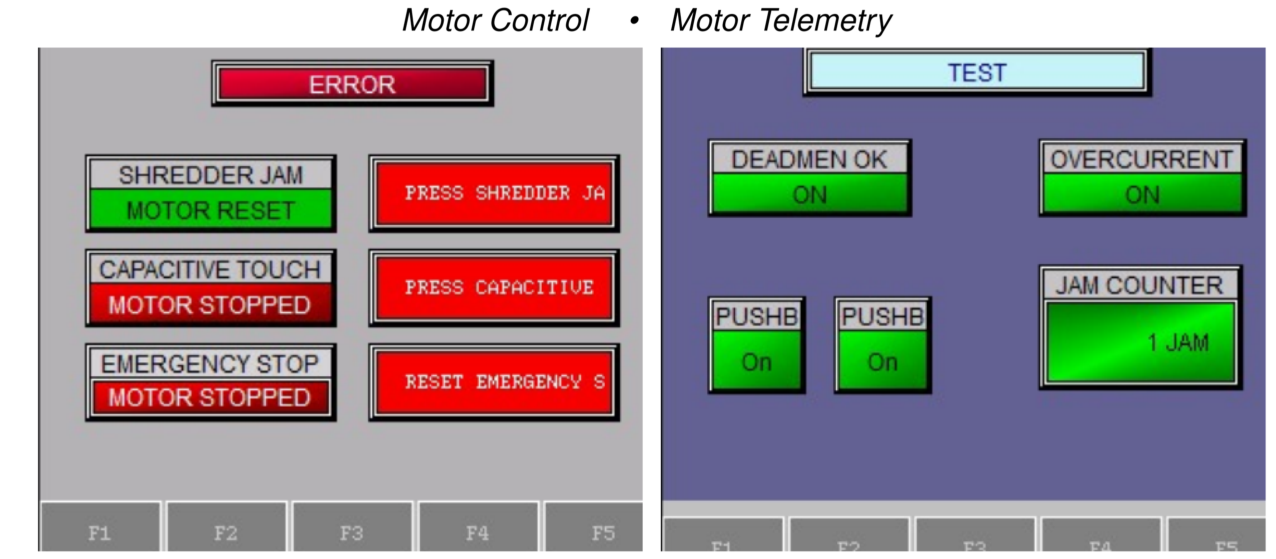
Jam-recovery rungs (Page 4 of 7): E-stop response, overcurrent latch, 2 s unjam reverse, and post-reverse coast-to-stop timing.

HMI

The operator interface is an AutomationDirect EA1-T4CL C-more Micro-Graphic 4 in. touch panel running `HMI.413.mgp`, serial to the H2-DM1E CPU. Four screens hold 27 PLC tags:



Motor Control • Motor Telemetry



Error / Alarm • Test / Commissioning

Safety & Compliance

- **E-stop:** Category 0 stop. Cutler-Hammer E22B1 NC mushroom in series with the SD-N35 contactor coil. Drops the contactor independently of the PLC.
- **Two-hand / deadman:** both pushbuttons must be held to enable the motor (PLC X2 AND X3, with HMI soft mirrors C23 / C24).
- **Overcurrent / jam:** VFD over-torque trip (`PD124 = 150%`, `PD125 = 3 s`) drives PLC X5 through the fault relay; ladder runs the 3-strike unjam routine.
- **Enclosure:** Vorksmen SEB001 steel, manufacturer-rated **NEMA Type 12** (dust-tight, drip-tight).
- **Compliance:** NFPA 70 (NEC), NFPA 79, NEMA 250 / ICS / MG 1, ISO 12100, ISO 13849-1 (preliminary Cat B–1 / PL b–c), ISO 13850, ISO 13851, OSHA 29 CFR 1910.147 (LOTO).

Results

- Panel built, wired, and bench-validated. All 31 ladder rungs program and scan correctly.
- All four HMI screens operate; tag database matches the PLC.
- Speed reference path verified: HMI `V2000` → `SCALE` → `F2-08DA-2 CH1` → VFD `VI`.
- Jam-recovery verified: overcurrent on `X5` latches `C10` → 2 s reverse (`T1`) → coast-to-stop buffer (`T3`).
- Three-revision point list reviewed by Fearghus Tyler at each stage.

Future Work

- Production motor selection (3–5 HP) supersedes `PD141--PD144`.
- Mechanical integration / shred-load run pending ME team.
- Capacitive-touch module on `X7`: machine-level verification pending.
- Optional Modbus link to VFD for richer telemetry.

Acknowledgments

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